

MPowerBoards — PPT Text

The Problem:

- Power systems are difficult to visualize
- Software and equations alone aren't enough
- Existing hardware is expensive or inaccessible

Every year, thousands of engineering students learn power systems through equations and software—but never truly see how the grid behaves. Existing teaching tools are either too abstract, too expensive, or inaccessible.

We introduce to you, Mpower Boards.

MPowerBoards: Interactive Physical Visualization

- Real-time simulations
- Embedded electronics
- Modeling complex power systems concepts

MPowerBoards transforms real power-grid simulations into interactive physical boards using LEDs, embedded electronics, and real-time visualization. Students don't just calculate power flow—they watch it happen.

We've developed three flagship products for classrooms, education, and research, allowing users to explore everything from basic concepts to real grid operations.

Progress / Validation:

- 15+ Undergraduate/ Highschool students
- 10+ K-12 Teachers and Students
- 8+ Professors and PhD researchers
- Working prototypes
- Classroom demos
- Georgia Tech faculty
- AI for Power Systems Institute

We've already built working prototypes and are collaborating with Georgia Tech faculty and the AI Institute to bring MPowerBoards into classrooms and outreach.

MPowerBoards

Revolutionizing Power Grid Education and Research

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Our mission is to make power systems education visual, interactive, and accessible. We'd love to connect with educators, researchers, and investors.

Progress Since Kickoff

Focusing on **3 main customers** and hence, **3 main flagship designs**:

Georgia Tech Prof. 1:

Wants a more decorative MPB for displaying Independent System Operator information, for *use in his office*.

Georgia Tech Prof. 2:

Wants a more realistic MPB (not of the U.S. but of an African country) for *use in his classes next semester*.

High School/ Undergraduate Students:

Partnered with a Georgia Tech AI Institution that has regular interactions with high schools and undergrads to present demos and record for advertisement.